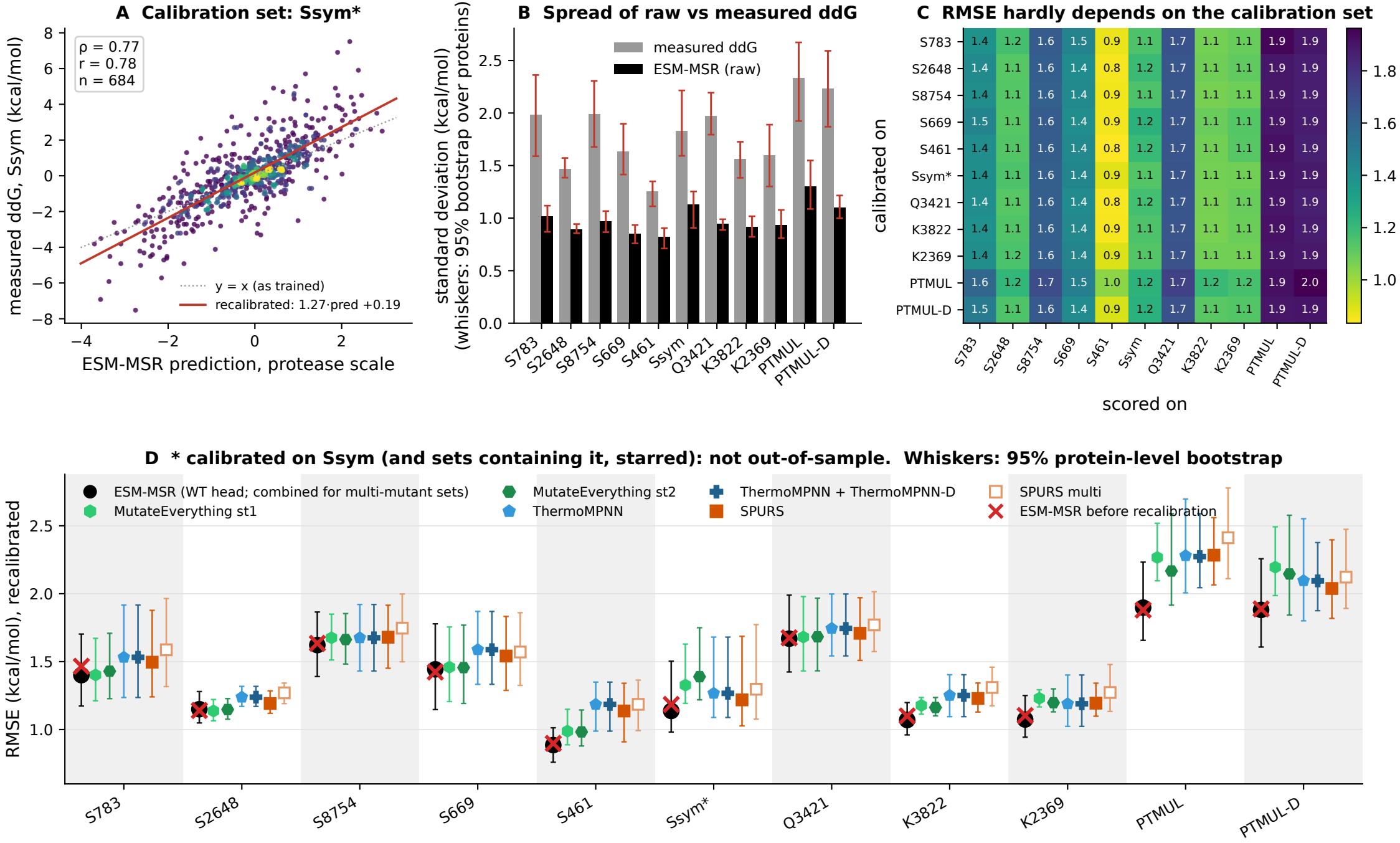

A Calibration set: Ssym*
ρ = 0.77
r = 0.78
n = 684
y = x (as trained)
recalibrated: 1.27·pred +0.19
measured ddG, Ssym (kcal/mol)
ESM-MSR prediction, protease scale
B Spread of raw vs measured ddG
measured ddG
ESM-MSR (raw)
standard deviation (kcal/mol)
(whiskers: 95% bootstrap over proteins)
S783 S2648 S8754 S669 S461 Ssym Q3421 K3822 K2369 PTMUL PTMUL-D
C RMSE hardly depends on the calibration set
calibrated on
S783 1.4 1.2 1.6 1.5 0.9 1.1 1.7 1.1 1.1 1.9 1.9
S2648 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
S8754 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
S669 1.5 1.1 1.6 1.4 0.9 1.2 1.7 1.1 1.1 1.9 1.9
S461 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
Ssym* 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
Q3421 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
K3822 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
K2369 1.4 1.2 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
PTMUL 1.6 1.2 1.7 1.5 1.0 1.2 1.7 1.2 1.2 1.9 2.0
PTMUL-D 1.5 1.1 1.6 1.4 0.9 1.2 1.7 1.1 1.1 1.9 1.9
scored on
D * calibrated on Ssym (and sets containing it, starred): not out-of-sample. Whiskers: 95% protein-level bootstrap
ESM-MSR (WT head; combined for multi-mutant sets)
MutateEverything st1
MutateEverything st2
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
SPURS
SPURS multi
ESM-MSR before recalibration
RMSE (kcal/mol), recalibrated
S783 S2648 S8754 S669 S461 Ssym* Q3421 K3822 K2369 PTMUL PTMUL-D